

WEEK 5 – HEALTHCARE DATA CLEANING AND DATA MODELING

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1. Introduction

The Week 5 project focuses on preparing and organizing healthcare data using Power BI Power Query. The raw healthcare dataset was cleaned, transformed and divided into three separate tables:

- ❖ Dim_Patient,
- ❖ Dim_Admission, &
- ❖ Fact_Billing.

This prepared dataset can be used for further analysis and visualization in Week 6.

2. Objective

The main objectives of this project are:

- To clean the raw healthcare dataset.
- To remove duplicate records.
- To remove unnecessary columns.
- To correct data types.
- To standardize text values.
- To create useful derived columns.
- To divide the dataset into three meaningful tables.
- To create relationships between the tables.

3. Dataset

The project uses a **Healthcare Dataset** containing patient, admission, medical and billing-related information.

The original dataset contains details such as:

- Patient ID
- Name
- Age
- Gender
- Blood Type
- Medical Condition
- Doctor
- Hospital
- Insurance Provider
- Admission Type
- Date of Admission
- Discharge Date
- Medication
- Test Results
- Billing Amount
- Room Number

4. Data Cleaning Using Power Query

The dataset was imported into **Power BI Power Query Editor** for cleaning and transformation.

4.1 Promoted Headers

The first row of the dataset was promoted as the column headers so that each field could be correctly identified.

4.2 Changed Data Types

Appropriate data types were assigned to the columns.

Examples:

- Age → Whole Number
- Room Number → Whole Number
- Billing Amount → Decimal Number
- Date of Admission → Date
- Discharge Date → Date
- Text fields → Text

4.3 Cleaned Text

The **Clean** transformation was applied to remove unwanted non-printing characters from text values.

4.4 Trimmed Text

The **Trim** transformation was used to remove unnecessary spaces from the beginning and end of text values.

4.5 Uppercase Text

Text values were standardized using the Uppercase transformation where required.

4.6 Removed Duplicates

Duplicate records were identified and removed to improve data quality and avoid repeated information.

4.7 Removed Unnecessary Columns

Columns that were not required for the analysis were removed from the respective tables.

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File Home Transform Add Column View Tools Help

Close & Apply New Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Text Use First Row as Headers Merge Queries Append Queries Combine Files Combine

Queries [4] healthcare_dataset Dim_Patient Dim_Admission Fact_Billing

fx = Table.AddIndexColumn(#"Renamed Columns1", "Index", 1, 1, Int64.Type)

	Name	Age	Gender	Blood Type	Medical Condition
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
1	BOBBY JACKSON	30	Male	B-	Cancer
2	LESLIE TERRY	62	Male	A+	Obesity
3	DANNY SMITH	76	Female	A-	Obesity
4	ANDREW WATTS	28	Female	O+	Diabetes
5	ADRIENNE BELL	43	Female	AB+	Cancer
6	EMILY JOHNSON	36	Male	A+	Asthma
7	EDWARD EDWARDS	21	Female	AB-	Diabetes
8	CHRISTINA MARTINEZ	20	Female	A+	Cancer
9	JASMINE AGUILAR	82	Male	AB+	Asthma
10	CHRISTOPHER BERG	58	Female	AB-	Cancer
11	MICHELLE DANIELS	72	Male	O+	Cancer
12	AARON MARTINEZ	38	Female	A-	Hypertension
13	CONNOR HANSEN	75	Female	A+	Diabetes
14	ROBERT BAUER	68	Female	AB+	Asthma
15	BROOKE BRADY	44	Female	AB+	Cancer
16	MS. NATALIE GAMBLE	46	Female	AB-	Obesity
17	HALEY PERKINS	63	Female	A+	Arthritis
18	MRS. JAMIE CAMPBELL	38	Male	AB-	Obesity
19	LUKE BURGESS	34	Female	A-	Hypertension
20	DANIEL SCHMIDT	63	Male	B+	Asthma
21	TIMOTHY BURNS	67	Female	A-	Asthma
22	CHRISTOPHER BAUER	40	Male	B-	Asthma

Query Settings

PROPERTIES

Name healthcare_dataset

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Uppercased Text
- Cleaned Text
- Trimmed Text
- Changed Type1
- Added Conditional Column
- Added Index
- Renamed Columns
- Removed Columns
- Added Index1
- Renamed Columns1
- Added Index2

5. Creation of Tables

After cleaning the data, the dataset was divided into three tables.

5.1 Dim Patient

The **Dim_Patient** table contains patient-related information.

Columns:

- Patient ID
- Name
- Gender
- Age
- Blood Type
- Age Group

The **Age Group** column was created using a conditional column based on patient age.

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File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Remove Keep Remove Split Group Data Type: Any Merge Queries Append Queries Combine Files

Queries [3] healthcare_dataset Dim_Patient Dim_Admission

Table.Distinct(#"Removed Columns1")

	Gender	Name	Age	Blood Type	Age Group
1	Male	BOBBY JACKSON	30	B-	Younger Adult
2	Male	LESLIE TERRY	62	A+	Elder Adult
3	Female	DANNY SMITH	76	A-	Elder Adult
4	Female	ANDREW WATTS	28	O+	Younger Adult
5	Female	ADRIENNE BELL	43	AB+	Adult
6	Male	EMILY JOHNSON	36	A+	Younger Adult
7	Female	EDWARD EDWARDS	21	AB-	Younger Adult
8	Female	CHRISTINA MARTINEZ	20	A+	Younger Adult
9	Male	JASMINE AGUILAR	82	AB+	Elder Adult
10	Female	CHRISTOPHER BERG	58	AB-	Adult
11	Male	MICHELLE DANIELS	72	O+	Elder Adult
12	Female	AARON MARTINEZ	38	A-	Younger Adult
13	Female	CONNOR HANSEN	75	A+	Elder Adult
14	Female	ROBERT BAUER	68	AB+	Elder Adult
15	Female	BROOKE BRADY	44	AB+	Adult
16	Female	MS. NATALIE GAMBLE	46	AB-	Adult
17	Female	HALEY PERKINS	63	A-	Elder Adult
18	Female	LUKE BURGESS	34	A-	Younger Adult
19	Female	TIMOTHY BURNS	67	A-	Elder Adult
20	Male	CHRISTOPHER BRIGHT	48	B+	Adult
21	Male	DR. EILEEN THOMPSON	59	A+	Adult
22	Male	REYD ELLERMAN	73	AB-	Elder Adult

Query Settings

PROPERTIES

Name Dim_Patient

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Uppercased Text

Cleaned Text

Trimmed Text

Changed Type1

Added Conditional Column

Added Index

Renamed Columns

Removed Duplicates

Removed Duplicates1

Removed Other Columns

Removed Duplicates2

Removed Columns

Added Index1

Renamed Columns1

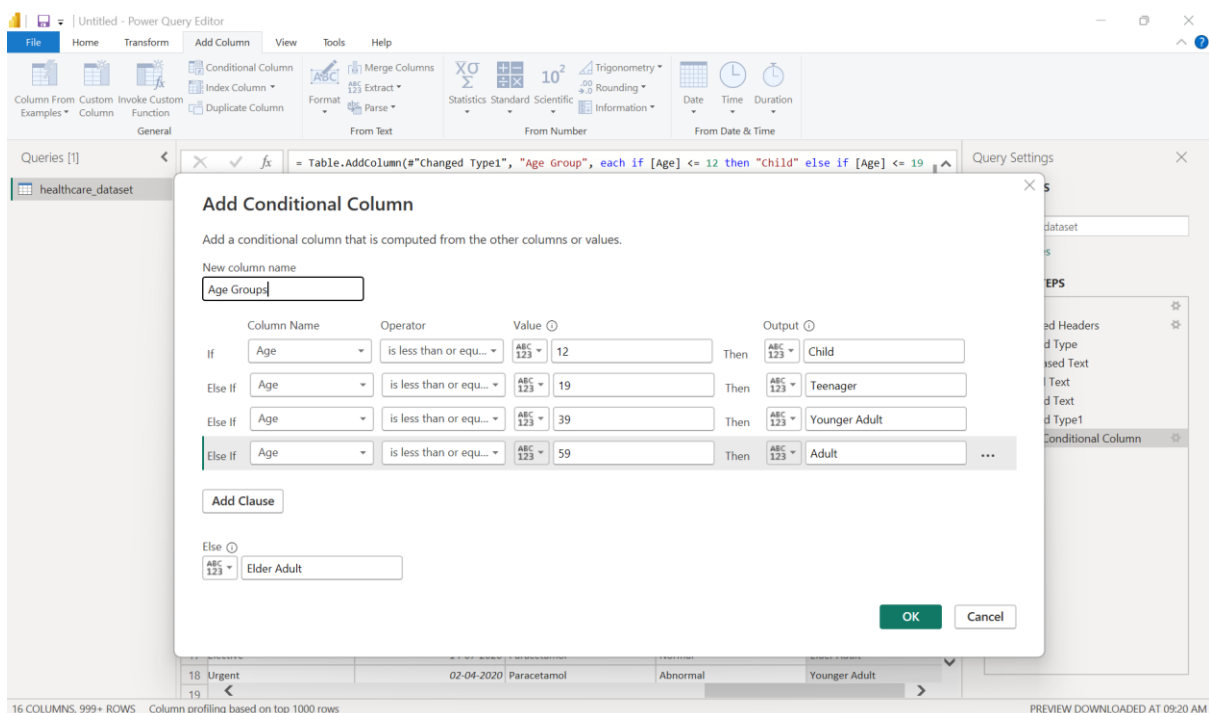
Removed Duplicates3

Removed Columns1

Removed Duplicates4

Age groups used:

- Child
- Teenager
- Younger Adult
- Adult
- Elder Adult



5.2 Dim_Admission

The **Dim_Admission** table contains hospital admission and treatment-related information.

Columns:

- Admission ID
- Admission Type
- Doctor

- Hospital
- Medical Condition
- Medication
- Room Number
- Test Results

This table helps organize information related to each hospital admission.

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File Home Transform View Tools Help

Column From Custom Invoke Custom Examples Column Function General

Conditional Column Index Column Duplicate Column

Format Merge Columns Extract Parse From Text

Statistics Standard Scientific Trigonometry Rounding Information

Date Time Duration From Date & Time

Queries [4]

healthcare_dataset Dim_Patient Dim_Admission Fact_Billing

fx = Table.Distinct(#"Renamed Columns2")

	Medical Condition	Doctor	Hospital	Room Number	Admission Type
1	Cancer	Matthew Smith	Sons and Miller	328	Urgent
2	Obesity	Samantha Davies	Kim Inc	265	Emergency
3	Obesity	Tiffany Mitchell	Cook PLC	205	Emergency
4	Diabetes	Kevin Wells	Hernandez Rogers and Vang,	450	Elective
5	Cancer	Kathleen Hanna	White-White	458	Urgent
6	Asthma	Taylor Newton	Nunez-Humphrey	389	Urgent
7	Diabetes	Kelly Olson	Group Middleton	389	Emergency
8	Cancer	Suzanne Thomas	Powell Robinson and Valdez,	277	Emergency
9	Asthma	Daniel Ferguson	Sons Rich and	316	Elective
10	Cancer	Heather Day	Padilla-Walker	249	Elective
11	Cancer	John Duncan	Schaefer-Porter	394	Urgent
12	Hypertension	Douglas Mayo	Lyons-Blair	288	Urgent
13	Diabetes	Kenneth Fletcher	Powers Miller, and Flores	134	Emergency
14	Asthma	Theresa Freeman	Rivera-Gutierrez	309	Urgent
15	Cancer	Roberta Stewart	Morris-Arellano	182	Urgent
16	Obesity	Maria Dougherty	Cline-Williams	465	Elective
17	Arthritis	Erica Spencer	Cervantes-Wells	114	Elective
18	Obesity	Justin Kim	Torres, and Harrison Jones	449	Urgent
19	Hypertension	Justin Moore Jr.	Houston PLC	260	Elective
20	Asthma	Denise Galloway	Hammond Ltd	465	Elective
21	Asthma	Krista Smith	Jones LLC	115	Elective
22	Asthma	Gregory Smith	Wolfe and Sons	306	Urgent

Query Settings

PROPERTIES

Name: Dim_Admission

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Uppercased Text
- Cleaned Text
- Trimmed Text
- Changed Type1
- Added Conditional Column
- Added Index
- Renamed Columns
- Added Index1
- Renamed Columns1
- Removed Other Columns
- Removed Duplicates
- Removed Columns
- Removed Duplicates1
- Added Index2
- Renamed Columns2
- Removed Duplicates2

5.3 Fact Billing

The **Fact_Billing** table contains billing and date-related information.

Columns:

- Billing Amount
- Date of Admission

- Discharge Date
- Insurance Provider
- Admission ID
- Patient ID

This table acts as the main fact table for future billing and healthcare analysis.

	Dim_Admission.Admission ID	Dim_Patient.Patient ID	Billing Amount	Insurance Provider	Date of Adm
1	1	1	18,856.28	Blue Cross	
2	2	2	33,643.33	Medicare	
3	3	3	27,955.10	Aetna	
4	4	4	37,909.78	Medicare	
5	5	5	14,238.32	Aetna	
6	6	6	48,145.11	UnitedHealthcare	
7	7	7	19,580.87	Medicare	
8	8	8	45,820.46	Cigna	
9	9	9	50,119.22	Cigna	
10	10	10	19,784.63	UnitedHealthcare	
11	11	11	12,576.80	Medicare	
12	12	12	7,999.59	Medicare	
13	13	13	43,282.28	Cigna	
14	14	14	33,207.71	UnitedHealthcare	
15	15	15	40,701.60	UnitedHealthcare	
16	16	16	12,263.36	Blue Cross	
17	17	17	24,499.85	UnitedHealthcare	
18	18	18	17,440.47	Cigna	
19	19	19	18,843.02	Blue Cross	
20	20	20	23,762.20	Cigna	
21	21	21	42.51	Blue Cross	
22	22	22	17,605.01	Aetna	

6. Data Modeling

After creating the three tables, relationships were created between the tables.

The model consists of:

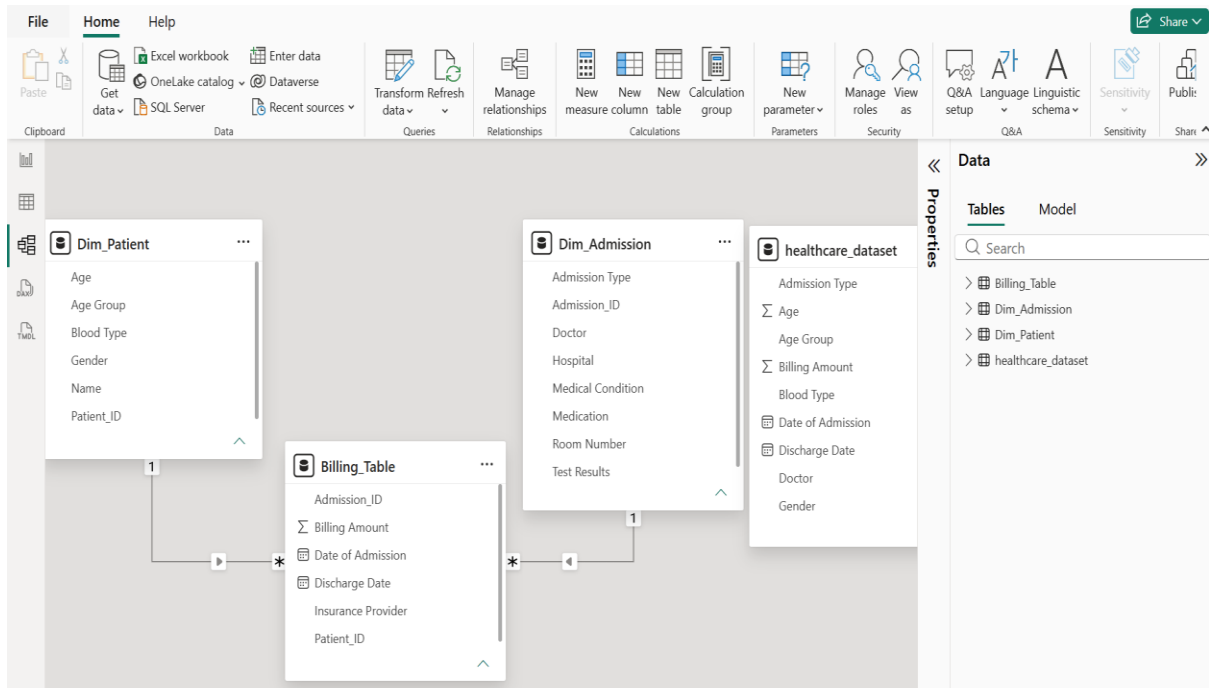
Dim_Patient → Fact_Billing

and

Dim_Admission → Fact_Billing

The **Patient ID** and **Admission ID** are used as connecting fields.

This structure separates descriptive information from transaction/billing information and makes the dataset easier to analyze.



7. Data Quality Improvements

The following improvements were made during the cleaning process:

Data Quality Issue

Unwanted spaces

Unwanted characters

Inconsistent text format

Incorrect data types

Action Taken

Trimmed

Cleaned

Standardized

Changed

Data Quality Issue	Action Taken
Duplicate records	Removed
Unnecessary columns	Removed
Age classification	Age Group created
Large raw dataset	Divided into 3 tables

8. Final Data Model

The final project contains three cleaned tables:

Dim_Patient

Patient demographic information.

Dim_Admission

Admission and medical information.

Fact_Billing

Billing and date information.

The three-table structure provides an organized foundation for the next stage of the project.

9. Conclusion

In Week 5, the raw healthcare dataset was successfully cleaned and transformed using **Power Query**. Unnecessary columns and duplicate records were removed, text values were cleaned and trimmed, and appropriate data types were assigned. A new **Age Group** column was also created.

The final dataset was organized into **Dim_Patient**, **Dim_Admission** and **Fact_Billing** tables and connected through relationships. This cleaned and structured data model is now ready for **Week 6 visualization and chart creation**.